Proofs to the theoretical study

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1 Proofs in the theoretical analysis

Theorem 1 (Weak Submodularity of FIGA). *Let $\mathcal{G} = (\mathcal{V}_{fl} \cup \mathcal{V}_{ep}, \mathcal{E})$ be a heterogeneous graph representation of network traffic. Subject to additive-only and endpoint reachability constraints ($\mathcal{V}_{ep}^{dst} \subseteq \mathcal{V}_{ep}$), let $\mathcal{S}_{inj} = \{(v_{ep}^{(i)*}, \mathbf{x}_{fl}^{(i)*})\}_{i=1}^{K}$ be a set of injected cover flows within an attack budget K. The adversary, controlling endpoint $v_{ep,atk} \in \mathcal{V}_{ep}$, seeks to evade detection for a target malicious flow node $v_{fl,m} \in \mathcal{V}_{fl}$.*

Let $m = AGGREGATE(\{\mathbf{h}_u : u \in \mathcal{N}(v_{fl,m})\})$ be the mean-pooling based aggregated neighborhood message at node $v_{fl,m}$, which serves as the continuous latent representation used for classification. Let $\ell_{sur}(m)$ and $\ell_{tar}(m)$ denote the evasion objectives for the surrogate ($g_{\mathcal{S}}$) and target ($g_{\mathcal{T}}$) models, respectively. Assume ℓ_{sur} is L-Lipschitz and μ-strongly concave with respect to m. If the cross-model gradient alignment is bounded such that for any message shift Δm induced by a single injection, $|\langle \nabla \ell_{tar}(m) - \nabla \ell_{sur}(m), \Delta m \rangle|$ $\leq \epsilon$, then maximizing the surrogate objective $\ell_{sur}(m(\mathcal{S}_{inj}))$ is a γ_{sur}-weakly submodular maximization problem [1]. Furthermore, the target evasion loss is bounded by:

$$\ell_{tar}(m(\mathcal{S}_{inj})) \geq \ell_{sur}(m(\mathcal{S}_{inj})) - K\epsilon,$$

ensuring that maximizing the surrogate evasion loss effectively maximizes a strict lower bound of the target model's evasion objective.

Proof. The proof consists of two components: establishing the weak submodularity of the surrogate evasion objective over the discrete graph domain, and deriving the strict transferability lower bound for the target NIDS.

Part 1: Weak Submodularity of the Surrogate Objective.
Let $F_{sur}(\mathcal{S}) = \ell_{sur}(m(\mathcal{S}))$ define the discrete set-function representing the surrogate evasion objective, evaluated over a chosen subset of flow injections $\mathcal{S} \subseteq \mathcal{Z}$, where $\mathcal{Z}$ is the discrete pool of valid injection candidate pairs (endpoint and feature vector) and $|\mathcal{S}| \leq K$.

The target flow node $v_{fl,m}$ aggregates information from its neighborhood $\mathcal{N}(v_{fl,m})$. Under the mean-pooling aggregation mechanism assumed in the theorem, the continuous latent representation $m(\mathcal{S})$ is defined as:

$$m(\mathcal{S}) = \frac{1}{|\mathcal{N}(v_{fl,m})| + |\mathcal{S}_{vic}|} \sum_{u \in \mathcal{N}(v_{fl,m}) \cup \mathcal{S}_{vic}} \mathbf{h}_u \tag{1}$$

where $\mathcal{S}_{vic}$ represents the set of injected flows that directly or indirectly route to the victim's neighborhood. When a new covert flow f_{cov} is injected, the marginal shift in the aggregated message space $\Delta m = m(\mathcal{S} \cup \{f_{cov}\}) - m(\mathcal{S})$ strictly and monotonically decays due to the structural degree normalization factor.

By the theorem's assumption, the continuous surrogate objective $\ell_{sur}(m)$ is L-Lipschitz and μ-strongly concave. Because the structural mapping $m(\mathcal{S})$ exhibits diminishing incremental shifts, Elenberg et al. [1] established that optimizing an L-Lipschitz, μ-strongly concave function subject to a discrete cardinality constraint satisfies Restricted Strong Concavity (RSC) and Restricted Lipschitzness (RSM).

Consequently, the discrete set-function $F_{sur}(\mathcal{S})$ inherits weak submodularity. Specifically, there exists a weak submodularity ratio $\gamma_{sur} \in (0, 1]$, bounded strictly below by the condition number of the objective ($\gamma_{sur} \geq \mu/L$), such that for any disjoint sets $\mathcal{S}$ and $\mathcal{T}$:

$$F_{sur}(\mathcal{S} \cup \mathcal{T}) - F_{sur}(\mathcal{S}) \geq \gamma_{sur} \sum_{f_{cov} \in \mathcal{T}} [F_{sur}(\mathcal{S} \cup \{f_{cov}\}) - F_{sur}(\mathcal{S})] \tag{2}$$

This formally establishes that maximizing $F_{sur}(\mathcal{S})$ via sequential flow injection is a γ_{sur}-weakly submodular maximization problem.

Part 2: Black-Box Target Transfer Bound.
To establish transferability to the target NIDS ($g_{\mathcal{T}}$), we map the target set-function $F_{tar}(\mathcal{S}) = \ell_{tar}(m(\mathcal{S}))$ to the surrogate space.

Let $\mathcal{S}_{inj} = \{z_1, \ldots, z_K\}$ be the sequential set of injected flows, and let Δm_k denote the message shift induced by the k-th injection. By the first-order Taylor expansion of the L-Lipschitz objectives, the marginal gain of adding the k-th covert flow is tightly bounded by its directional derivative:

$$F_{tar}(z_1, \ldots, z_k) - F_{tar}(z_1, \ldots, z_{k-1}) \approx \langle \nabla \ell_{tar}(m_{k-1}), \Delta m_k \rangle \tag{3}$$

The theorem explicitly bounds the cross-model gradient alignment such that the proxy gap for any single valid injection shift satisfies $|\langle \nabla \ell_{tar} - \nabla \ell_{sur}, \Delta m_k \rangle| \leq \epsilon$. Substituting this boundary condition, we relate the marginal gains across the two models:

$$F_{tar}(z_1, \ldots, z_k) - F_{tar}(z_1, \ldots, z_{k-1}) \geq (F_{sur}(z_1, \ldots, z_k) - F_{sur}(z_1, \ldots, z_{k-1})) - \epsilon \tag{4}$$

Summing this inequality telescopically over the entire attack budget K:

$$\sum_{k=1}^{K} \Delta F_{tar}^{(k)} \geq \sum_{k=1}^{K} \left[\Delta F_{sur}^{(k)} - \epsilon \right] \tag{5}$$

Which yields the final total transferable gain:

$$\ell_{tar}(m(\mathcal{S}_{inj})) - \ell_{tar}(m_0) \geq \ell_{sur}(m(\mathcal{S}_{inj})) - \ell_{sur}(m_0) - K\epsilon \tag{6}$$

Assuming initial alignment of the baseline loss prior to the attack, this proves the strict lower bound $\ell_{tar}(m(\mathcal{S}_{inj})) \geq \ell_{sur}(m(\mathcal{S}_{inj})) - K\epsilon$, guaranteeing that maximizing the surrogate objective yields bounded, predictable evasion success on the live target system.

2 Proof of Corollary 1 (Optimality Guarantee and Convergence)

Corollary 1 (Optimality Guarantee of the Influence-Function Search). *Let $\mathcal{I}(v_{fl}, g_{\mathcal{S}})$ be the influence function approximating the marginal gain $\ell_{sur}(\mathcal{T} \cup \{v_{fl}\}) - \ell_{sur}(\mathcal{T})$ without computing full GNN forward-backward passes. The proposed FIGA method, as given in Algorithm 1, acts as a discrete greedy maximization algorithm. Selecting the injected set $\hat{\mathcal{S}}_K$ iteratively provides a $(1 - e^{-\gamma_{sur}})$-approximation of the optimal surrogate attack $\mathcal{O}^*_{sur}$, converging in polynomial time $\mathcal{O}(K \cdot |\mathcal{V}^{pool}_{fl}| \cdot \mathcal{C}_{inf})$, where $\mathcal{C}_{inf}$ is the computational complexity of the influence score.*

Proof. The proof relies on classical results for the greedy maximization of weakly submodular set-functions subject to cardinality constraints.

By Theorem 1, we established that the surrogate evasion objective $F_{sur}(\mathcal{S}) = \ell_{sur}(m(\mathcal{S}))$ is a γ_{sur}-weakly submodular, monotonically non-decreasing set-function over the discrete injection space $\mathcal{Z}$, where the submodularity ratio $\gamma_{sur} \in (0, 1]$.

The proposed attack (Algorithm 1) operates as a discrete greedy algorithm. At each step $t \in \{1, \ldots, K\}$, it evaluates candidate injections using the influence function $\mathcal{I}(v_{fl}, g_{\mathcal{S}})$, which acts as a computationally efficient proxy for the true marginal gain:

$$\mathcal{I}(v_{fl}, g_{\mathcal{S}}) \approx F_{sur}(\mathcal{S}^{(t-1)} \cup \{v_{fl}\}) - F_{sur}(\mathcal{S}^{(t-1)}) \tag{7}$$

Assuming the influence function provides a sufficiently accurate first-order approximation of the true objective gradient, selecting the candidate that maximizes $\mathcal{I}$ is equivalent to performing a standard greedy selection step.

For the maximization of a monotonically non-decreasing, γ-weakly submodular function subject to a cardinality constraint K, the standard greedy algorithm produces a set $\hat{\mathcal{S}}_K$ that guarantees a strict approximation bound relative to the optimal set $\mathcal{O}^*_{sur}$ [1]. Specifically:

$$F_{sur}(\hat{\mathcal{S}}_K) \geq \left(1 - e^{-\gamma_{sur}}\right) F_{sur}(\mathcal{O}^*_{sur}) \tag{8}$$

This formally guarantees that the iterative influence-guided search identifies a near-optimal adversarial perturbation.

Polynomial Time Convergence: To construct the set $\hat{\mathcal{S}}_K$, the algorithm performs exactly K greedy iterations. During each iteration t, it must evaluate the influence function for every candidate valid flow in the search pool. Let the size of this valid search pool be denoted by $|\mathcal{V}^{pool}_{fl}|$, representing the combinations of available endpoints and benign features.

For each candidate, calculating the influence function $\mathcal{I}$ requires $\mathcal{C}_{inf}$ operations, avoiding the prohibitive cost of a full GNN forward-backward pass per candidate. Thus, the total computational cost per iteration is $|\mathcal{V}^{pool}_{fl}| \cdot \mathcal{C}_{inf}$.

Summing this cost over the K budgeted injections yields the total time complexity:

$$\mathcal{O}\left(K \cdot |\mathcal{V}^{pool}_{fl}| \cdot \mathcal{C}_{inf}\right) \tag{9}$$

Since K, $|\mathcal{V}_{fl}^{pool}|$, and $\mathcal{C}_{inf}$ are strictly bounded polynomials with respect to the graph size and model parameters, the algorithm is guaranteed to converge in polynomial time.

3 Proof of Proposition 1 (Query Complexity of Query-Based Attack Baselines)

Proof. Let the discrete search space of candidate perturbations at greedy step t be denoted as $\mathcal{Z} = \mathcal{V}_{ep}^{dst} \times \mathcal{X}_{val}$. The cardinality of this search space is exactly $N = |\mathcal{V}_{ep}^{dst}| \cdot |\mathcal{X}_{val}|$.

Under the hard-label black-box constraint, the target GNN model $\mathcal{T}$ acts as a binary oracle. For any candidate perturbation $z = (v_{ep}, \mathbf{x}_{fl}) \in \mathcal{Z}$, the target NIDS returns only a binary class label $y \in \{0, 1\}$ (benign or malicious), rather than continuous confidence scores or logits. Consequently, the marginal evasion gain $\ell_{evasion}\left(\mathcal{G}^{(t-1)} \oplus z\right)$ reduces to evaluating a black-box boolean function $f : \mathcal{Z} \to \{0, 1\}$.

The attacker's objective at step t is to identify the perturbation z^* that maximizes this marginal gain, or equivalently, to find a z such that $f(z) = 1$ (indicating a successful evasion shift).

In standard continuous domains (e.g., image perturbations), zeroth-order methods (such as ZO-SGD) estimate gradients using finite differences, achieving sub-linear query complexity relative to the search space size due to the inherent Lipschitzness and continuity of the feature space. However, in our problem-space setting, the topological injection of flows into $\mathcal{V}_{ep}^{dst}$ is strictly discrete. Furthermore, the feature validity constraint limits $\mathbf{x}_{fl}$ to a discrete pool $\mathcal{X}_{val}$.

Because the topological search space $\mathcal{Z}$ is discrete and the oracle $f(z)$ provides no directional gradient information (only hard labels), the function $f(z)$ lacks any exploitable Lipschitzness, convexity, or Lipschitz continuity with respect to the graph structure.

Thus, identifying z^* mathematically reduces to the problem of finding the maximum of a black-box boolean function over an unstructured discrete set. By standard information-theoretic lower bounds for randomized search (Yao's Minimax Principle for unstructured search [2]), any deterministic or randomized algorithm requires in the worst case an expected number of queries proportional to the size of the search space to find the target element with a constant probability of success.

Therefore, the number of queries $Q^{(t)}$ required to estimate the optimal injection z^* with constant accuracy is strictly bounded below by the cardinality of the search space:

$$Q^{(t)} \geq \Omega\left(|\mathcal{V}_{ep}^{dst}| \cdot |\mathcal{X}_{val}|\right) \tag{10}$$

This concludes the proof.

References

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